反常积分

右区间无限

定义

$$f(x) \in C[a,+\infty), \int_a^{+\infty} f(x)dx$$

$$\lim_{b\to\infty}[F(b)-F(a)] = A \Rightarrow \int_a^{+\infty} f(x)dx = A$$

$$\lim_{b\to\infty}[F(b)-F(a)]\text{不存在} \Rightarrow \int_a^{+\infty} f(x)dx\text{发散}$$

$$\int_1^{+\infty} \frac{dx}{\sqrt{x}(1+x)}$$

$$\forall b>1, \int_1^b \frac{dx}{\sqrt{x}(1+x)} = 2\int_1^b \frac{d\sqrt{x}}{1+(\sqrt{x})^2} = 2\arctan\sqrt{x}\mid_1^b = 2(\arctan\sqrt{b}-\frac{\pi}{4})$$

$$\because \lim_{b\to+\infty} 2(\arctan\sqrt{b}-\frac{\pi}{4}) = \frac{\pi}{2}$$

$$\therefore \int_1^{+\infty} \frac{dx}{\sqrt{x}(1+x)} = \frac{\pi}{2}$$

判别法

$$\text{若}\exists\alpha>1: \lim_{x\to+\infty} x^\alpha f(x)\text{存在} \Rightarrow \text{收敛}$$

$$\text{若}\exists\alpha\le 1: \lim_{x\to+\infty} x^\alpha f(x) \begin{cases} = A \neq 0 \\ = \infty \end{cases} \Rightarrow \text{发散}$$

$$\int_0^{+\infty} \frac{\sqrt{x}dx}{4+x^2}$$

$$\because \lim_{x\to\infty} x^{\frac{3}{2}} * \frac{\sqrt{x}}{4+x^2} = 1\text{且}\alpha = \frac{3}{2} > 1$$

$$\therefore \int_0^{+\infty} \frac{\sqrt{x}dx}{4+x^2}\text{收敛}$$

左区间无限

定义

$$f(x) \in C(-\infty,a], \int_{-\infty}^a f(x)dx$$

$$\forall b<a, \int_b^a f(x)dx = F(a)-F(b)$$

$$\text{若}\lim_{b\to-\infty}[F(a)-F(b)] = A \Rightarrow \int_{-\infty}^a f(x)dx = A$$

$$\text{若}\lim_{b\to-\infty}[F(a)-F(b)]\text{不存在} \Rightarrow \int_{-\infty}^a f(x)dx\text{发散}$$

判别法

$$若\exists\alpha>1:\lim_{x\to-\infty}x^{\alpha}f(x)存在\Rightarrow 收敛$$

$$若\exists\alpha\le 1:\lim_{x\to-\infty}x^{\alpha}f(x)\begin{cases}=A\neq 0\\=\infty\end{cases}\Rightarrow 发散$$

Gamma函数

$$\Gamma(\alpha)=\int_0^{+\infty}x^{\alpha-1}e^{-x}dx$$

$$\int_0^{+\infty}x\sqrt{x}e^{-x}dx=\int_0^{+\infty}x^{\frac{3}{2}}e^{-x}dx=\Gamma(\frac{5}{2})$$

$$\begin{cases}\Gamma(\alpha+1)=\alpha\Gamma(\alpha)\\\Gamma(n+1)=n!\\\Gamma(\frac{1}{2})=\sqrt{\pi}\end{cases}$$

$$\int_0^{+\infty}x^7e^{-x^2}dx=\frac{1}{2}\int_0^{+\infty}(x^2)^3e^{-x^2}dx^2=\frac{1}{2}\int_0^{+\infty}x^3e^{-x}dx=\frac{1}{2}\Gamma(4)=3$$

$$\int_0^{+\infty}x\sqrt{x}e^{-x}dx=\int_0^{+\infty}x^{\frac{3}{2}}e^{-x}dx=\Gamma(\frac{3}{2}+1)=\frac{3}{2}\Gamma(\frac{3}{2})=\frac{3}{2}*\frac{1}{2}*\Gamma(\frac{1}{2})=\frac{3\sqrt{\pi}}{4}$$

左端点无穷

定义

$$f(x)\in C(a,b]且f(a+0)=\infty,\int_a^b f(x)dx$$

$$\forall\epsilon>0,\int_{a+\epsilon}^b f(x)dx=F(b)-F(a+\epsilon)$$

$$\lim_{\epsilon\to0^+}[F(b)-F(a+\epsilon)]=A\Rightarrow\int_a^b f(x)dx=A$$

$$\lim_{\epsilon\to0^+}[F(b)-F(a+\epsilon)]不存在\Rightarrow\int_a^b f(x)dx发散$$

判别法

$$若\exists\alpha<1:\lim_{x\to a^+}(x-a)^{\alpha}f(x)存在\Rightarrow 收敛$$

$$若\exists\alpha\ge 1:\lim_{x\to a^+}(x-a)^{\alpha}f(x)\begin{cases}=A\neq 0\\=\infty\end{cases}\Rightarrow 发散$$

$$\int_0^{\frac{1}{2}}\frac{dx}{\sqrt{x(1-x)}}$$

$$\forall\epsilon>0,\int_{\epsilon}^{\frac{1}{2}}\frac{dx}{\sqrt{x(1-x)}}=2\int_{\epsilon}^{\frac{1}{2}}\frac{d\sqrt{x}}{\sqrt{1-(\sqrt{x})^2}}=2\arcsin\sqrt{x}\mid_{\epsilon}^{\frac{1}{2}}=2(\frac{\pi}{4}-\arcsin\sqrt{\epsilon})$$

$$\because\lim_{\epsilon\to0^+}2(\frac{\pi}{4}-\arcsin\sqrt{\epsilon})=\frac{\pi}{2}$$

$$\therefore\int_0^{\frac{1}{2}}\frac{dx}{\sqrt{x(1-x)}}=\frac{\pi}{2}$$

$$\int_0^1 \frac{dx}{\sqrt{x(x+1)}}$$

$$\lim_{x\to 0^+}(x-0)^{\frac{1}{2}}\frac{1}{\sqrt{x(x+1)}}=1\text{且}\alpha=\frac{1}{2}<1,\therefore\text{收敛}$$

$$\int_0^1 \frac{dx}{\sqrt{x(x+1)}}=2\int_0^1 \frac{d\sqrt{x}}{\sqrt{(\sqrt{x})^2+1}}=2\int_0^1 \frac{dx}{\sqrt{x^2+1}}=2\ln(x+\sqrt{x^2+1})\mid_0^1=2\ln(1+\sqrt{2})$$

右端点无穷

定义

$$f(x)\in C[a,b)\text{且}f(b-0)=\infty,\int_a^b f(x)dx$$

$$\forall\epsilon>0,\int_a^{b-\epsilon} f(x)dx=F(b-\epsilon)-F(a)$$

$$\lim_{\epsilon\to 0^+}[F(b-\epsilon)-F(a)]=A\Rightarrow\int_a^b f(x)dx=A$$

$$\lim_{\epsilon\to 0^+}[F(b-\epsilon)-F(a)]\text{不存在}\Rightarrow\int_a^b f(x)dx\text{发散}$$

判别法

$$\text{若}\exists\alpha<1:\lim_{x\to b^-}(b-x)^\alpha f(x)\text{存在}\Rightarrow\text{收敛}$$

$$\text{若}\exists\alpha\geq 1:\lim_{x\to b^-}(b-x)^\alpha f(x)\begin{cases}=A\neq 0\\=\infty\end{cases}\Rightarrow\text{发散}$$